ANITHA A

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PROFESSIONAL SUMMARY

Biomedical Engineering undergraduate passionate about the intersection of software and healthcare. Interested in developing technology-driven solutions using programming, IoT, and intelligent systems to improve healthcare accessibility and user experience. Continuously exploring new technologies and driven by a desire to learn, build, and contribute to meaningful innovations.

PROJECTS

Silver Nanoparticle integrated UV dye for smart textiles

- Developed an eco-friendly smart textile dye with enhanced UV sensitivity and antimicrobial properties.
- Contributed in the synthesis of silver nanoparticles , preparation of chitosan based formulations ,dye development and material characterisation to optimize the performance of smart textile applications.

Facial Recognition Attendance System

- Developed an attendance system using facial recognition and masked face detection for healthcare environments.
- Implemented user registration, identity verification, attendance tracking.

EchoThera – Breathing Guidance System

- Developed an IoT-based healthcare system for acquiring heart and lung sounds and enabling remote patient monitoring.
- Integrated biomedical sensors and embedded system components to acquire physiological signals and support real-time health monitoring
- Incorporated guided breathing assistance to improve patient usability and demonstrate an affordable healthcare monitoring solution.

ATHLETX – Sports Training App

- Developed an **AI-driven sports talent assessment application** that evaluates athletes using performance metrics to support objective talent identification and personalized training recommendations.
- Designed an intuitive mobile interface that enables athletes to record assessment data, monitor progress, and receive AI-based insights to improve performance over time.
- Built to promote **early talent identification**, reduce subjective bias in athlete evaluation, and provide coaches with a data-driven approach for tracking player development and selecting promising talent.

INTERNSHIPS

- **Ultraserve Systems , Chennai:** Gained exposure to ultrasound imaging systems, equipment operation, maintenance, and troubleshooting.
- **MedCuore Medical Solutions , Chennai:** Trained on ECG, EEG, EMG, BP, SpO2, and glucose monitoring devices with exposure to PCB components and sensor integration.

CERTIFICATIONS

1.Certification in NanoTechnology - TeachNook
2.Foundation Certificate in Palliative Care-1 - NPTEL (Score: 82%)
3.Introduction to Industry 4.0 and Industrial Internet of Things(Score: 78%)
4.Biophotonics - NPTEL(Score: 90%)
5.Healthcare Quality Management Infection Control and Patient Safety - NPTEL(Score: 80%)

TECHNICAL SKILLS

Programming Languages: Java, Python
Data & Analytics: NumPy, Pandas, Matplotlib, Power BI
Embedded Systems & IoT: ESP32, Arduino, Arduino IDE, Sensor Interfacing, IoT Systems .
Healthcare Technology & Prototyping: Wearable Device Prototyping, Healthcare Device Development, Sensor-Based System Design, Rapid Prototyping
Simulation & Engineering Tools: COMSOL Multiphysics, MATLAB.
Design & Creative Tools: Canva, Figma, Blender, Fusion 360, Presentation Design.

SOFT SKILLS

- Problem solving
- Analytical thinking
- Adaptability

EDUCATION

B.E. Biomedical Engineering – Rajalakshmi Engineering College, Chennai | CGPA: 7.79 | 2023–2027 **Higher Secondary Education** – The Children's Garden Higher Secondary School | 85.3%

ACHIEVEMENTS

• Top 27 Teams – EDII Hackathon, Government of Tamil Nadu.
• 2nd Prize – IPC Department-Level Project Competition for EchoThera.

LEADERSHIP & LANGUAGES

• Coordinator for a Clinical Engineering Workshop under the IEEE Student Society.
• Languages: Tamil, English, Hindi .